# Human Factors Engineering in Nuclear Safety

**“Human factors and nuclear safety since 1970 - A critical review of the past, present and future” (2021)**


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# Human factors and nuclear safety since 1970 – A critical review of the past, present and future

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ABSTRACT

Human Factors and Ergonomics (HFE) has played an essential role in increasing the safety and performance of the nuclear energy industry. This review aims to critically examine past HFE research on nuclear safety and provide insights into how HFE approaches can better address current challenges for nuclear safety and future changes (e.g. new technologies, Industry 4.0 and increasing system complexity). The research questions this paper addresses are: (1) what are the common HFE themes and topics that have been applied to nuclear safety? (2) what are the new/key emerging areas/ideas for nuclear safety from an HFE perspective? We reviewed (n = 334) articles from six prominent HFE and safety-focused journals. Articles were classified into a systems framework where research themes and topics could be classified and explored. The findings from the review indicate the primary topics which the nuclear energy industry have focused on have not changed much over time (e.g. safety culture; human error; human-machine interaction) and likewise methods from the 1970s continue to attract attention (e.g. probabilistic risk assessment and human reliability assessment). However, the review demonstrated that more recent topics such as resilience engineering and ideas centred on the 'Safety-II' and 'resilience engineering' movement are beginning to appear. One of the main conclusions from the review is that there is a need for greater exploration and experimentation of new and emerging areas of interest and their practical implications within the nuclear energy industry.

## 1. Introduction

The nuclear industry has proliferated rapidly in size in the post-war era. Nuclear energy currently provides 11% of the world's electricity from some 450 power reactors (World Nuclear Association, 2019a). Next-generation nuclear power plants (NPP) which incorporate standardised designs, new technology and increased safety are expected to be functional in the near future to fill the void caused by reduced fossil fuel usage and increased renewable energy production, e.g. 30% from renewables by 2020 (World Nuclear Association, 2019b). The nuclear industry is also considered a very prominent safety-critical industry. From the outset, there has been a keen awareness of the potential hazards of both nuclear criticality (uncontrolled nuclear fission chain reaction) and radioactive material release (World Nuclear Association, 2019c). As in other industries, the design, operation, governance, processes and procedures of nuclear power plants aims to minimise the likelihood of accidents and avoid human consequences when they occur (Stanton and Neville, 1996). However, there have been some high-profile disasters in the history of civil nuclear power: Three Mile Island (1979), Chernobyl (1986) and Fukushima (2011). The two most recent accidents achieved a maximum International Nuclear and Radiological Events Scale (INES) score of 7 signifying a considerable release of radioactive material with widespread health and environmental effects. In addition, there have been many near-miss events (Jones et al., 1999). However, the evidence over six decades shows that despite some high profile disasters, nuclear power is considered by many to be a 'safe' means of generating electricity and the risk of accidents in nuclear power plants is low and declining (Amalberti, 2000; Wheatley et al., 2016).

### 1.1. *Human factors and nuclear safety*

Over the last 40 years, Human Factors and Ergonomics (HFE) has played an essential role in increasing the safety of the nuclear industry. It has been applied in many different areas, e.g. the design and evaluation of processes, procedures, control rooms and accident analysis. One critical development that came out of the nuclear industry was the construct of safety culture, which was formalised as a result of the

# Overview

- Introduction to Human Factors Engineering (HFE) in Nuclear Safety
- Review of Nuclear Safety research since 1970
- Based on this literature review by Hamer, Waterson, and Jun in 2020.
- Explores how modern HFE research could better address modern and future challenges

# Objectives of the Literature Review

- Identify common HFE themes in Nuclear Power since 1970
- Identify new/key emerging areas for nuclear safety

# A Brief History of Nuclear Energy Developments and HFE

# Nuclear Energy

- Scientific research on atomic radiation began in 1895
- Research shifted from radiation to naval weapons and power in 1945
- Shifted again from weaponry to safe and reliable power plants in 1956

# Trends in HFE and Safety Science

- Human Reliability Analysis (HRA) became popular in the 1980s
- Cognitive psychology and decision-making research also became more widely investigated in HFE safety research around this time
- This produced the Cognitive Work Analysis and Cognitive Task Analysis
- The 1990s saw an increased investigation of the underlying causes of large-scale, complex system failures like Chernobyl. This marked a transition from just individual error to system error and safety culture.
- Our emerging "adaptive age" has seen resilience engineering (RE) and the efficiency-thoroughness trade-off principle, which are both proposing new ways of thinking about safety.

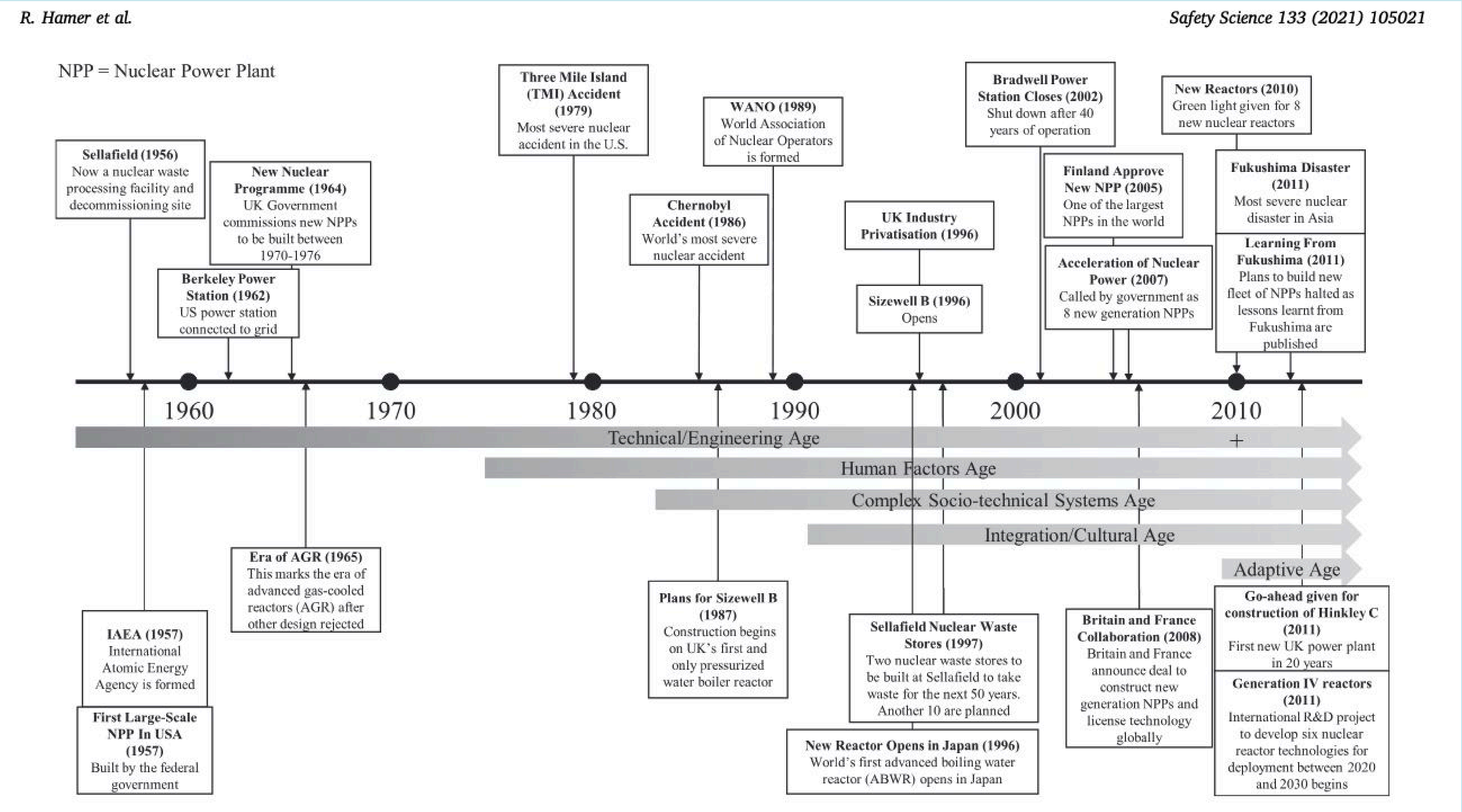



Fig. 1. Nuclear energy industry developments timeline.

# Introduction to Nuclear Safety

**Overview of nuclear energy and its safety-critical nature**

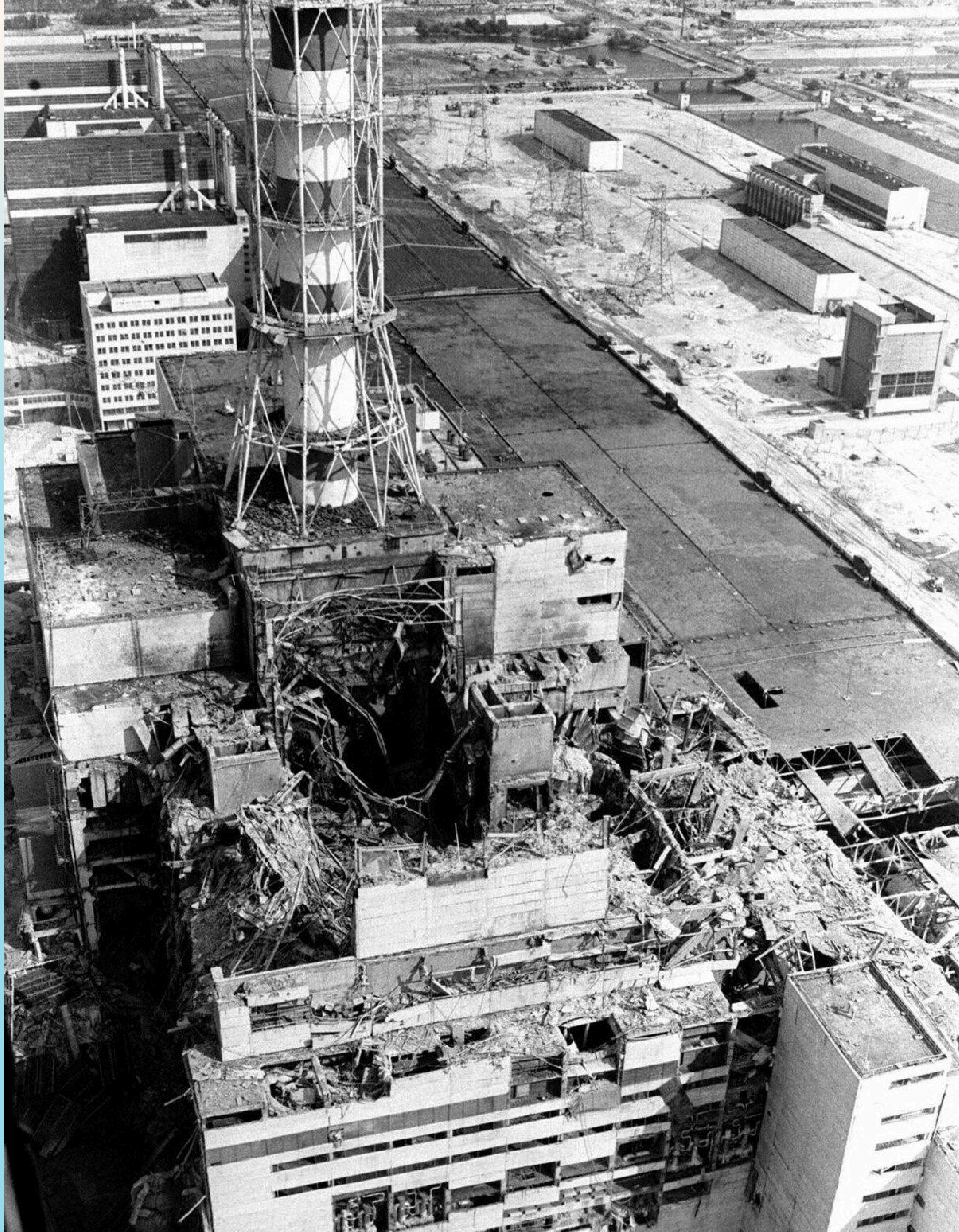
# Growth and Efficiency of Nuclear Energy

- Nuclear energy is expanding and becoming more efficient.
- As of 2019, there are 450 power reactors worldwide.
- Next-gen nuclear plants will feature standardized designs and new tech.
- These plants aim to replace fossil fuels and enhance safety in the near future.

# Prominent Safety-Critical Industry

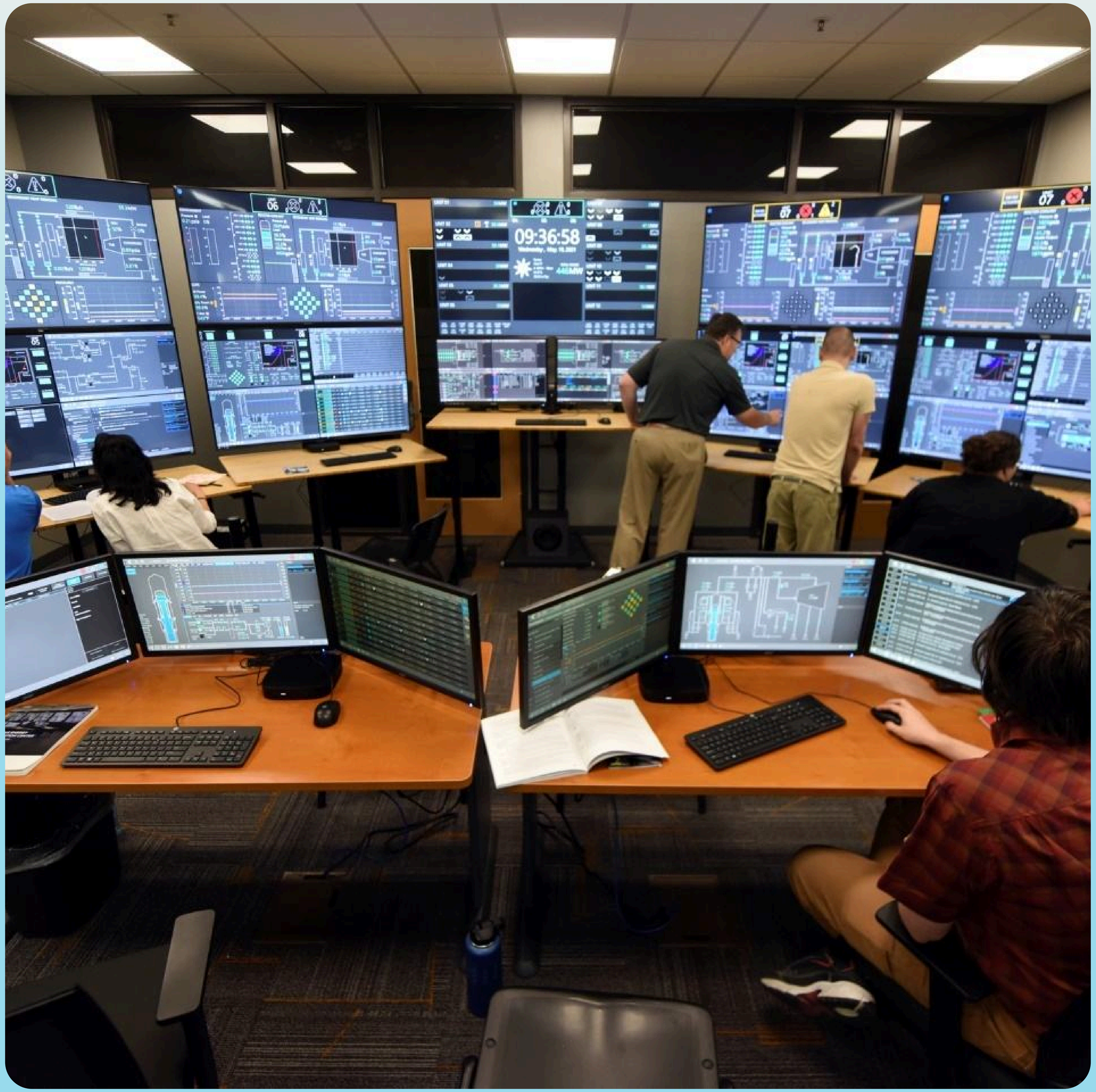
## High-Profile Nuclear Disasters

- Notable incidents: Three Mile Island (1979), Chernobyl (1986), Fukushima (2011).
- These events have heightened public concern about nuclear power.
- Despite these disasters, the risk of nuclear accidents is low.
- Safety measures have improved significantly over time.



# Human Factors Engineering (HFE) in Nuclear Safety

The role of HFE in enhancing nuclear safety




## HFE Applications in Nuclear Safety

- HFE is crucial for designing and evaluating processes, control rooms, and accident analysis.
- These analyses led to the formation of the construct “Safety Culture.”

# Safety Culture

- Safety Culture originated from the Chernobyl incident to improve safety management and was later adopted by other industries.
- Management and Organizational factors are critical for safety (not just the typical physical safety engineering operations).
- Overlaps with macro-ergonomics

# New Developments in Safety Management

Reliability Engineering and Safety-II

# Reliability Engineering (RE) and Safety-II

**Potential new research avenues for safety research and more proactive forms of safety management.**

| Safety-I | Safety-II |
|---|---|
| Traditional approach | New HFE approach |
| Focuses on identifying and eliminating hazards | Focuses on enhancing performance |
| Only observes how things go wrong | Observes why things go wrong + how they succeed |

The approach of Safety-II is especially helpful because success is often caused by the same factors that can lead to failures.

# Industry Developments (Industry 4.0) and HFE Opportunities

## (Hopefully, this means job security)

- Digital assistants can aid operator learning and performance.
- Cybersecurity is becoming increasingly important.
- Human-machine collaboration will become increasingly important in the age of automation and augmentation.
- Careful and systematic planning is required for decommissioning old nuclear power plants worldwide.
  - Defueling
  - Care and maintenance preparation
  - Site clearance
- HFE is considered essential for the design, integration, and evaluation of all NPP related equipment and safety cultures.

# Literature Review Results

# Results

## From 334 Relevant Articles in Six Journal Databases

- The most prominent category was ‘Methods, tools and instruments’ (40% of all articles).
- The second and third largest categories were ‘Organization’ (21% of all articles) and ‘individual’ (18% of all articles).
- The most prominent trends are the consistent increase of three categories: organization; individual; methods,  instruments and tools

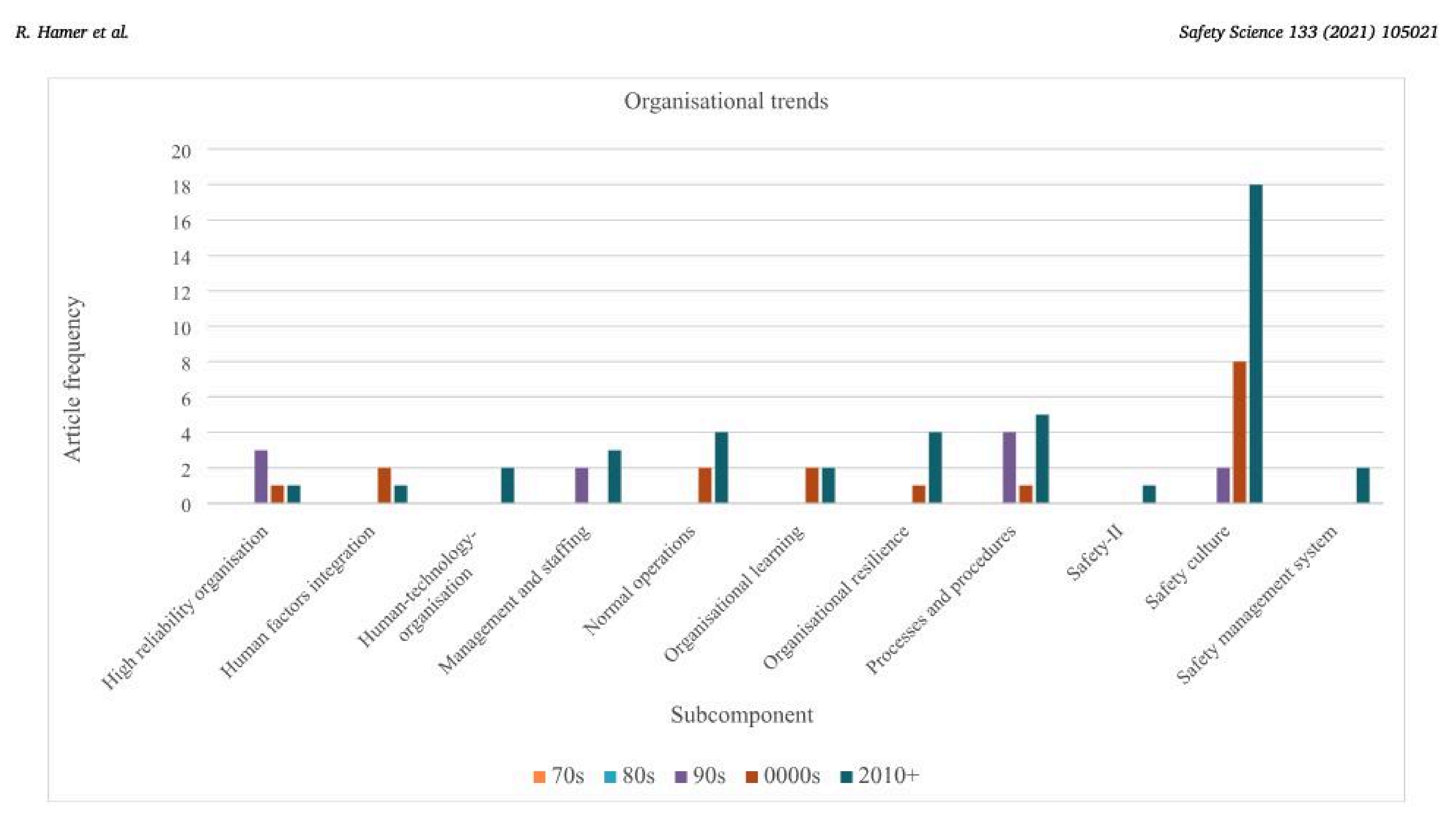



**Fig. 8.** Organisational trends.

## Exploration of Specific Trends

### Research that Increased:

- Human Error
- Human performance
  - Signal detection performance
  - Simulator trainings
  - Physical workload and shift work
  - Impact of role stressors and job dissatisfaction on risky behaviors
- Situational awareness (took off in the 2000s)
- Human Machine Interaction
- Cybersecurity
- Probabilistic Risk Assessment (PRA)
- Human Reliability Assessment (HRA)

### Research that stayed the same:

- Mental Workload
- Emergency operating procedures

# Discussion

Summary of Findings and Potential Developments

# Common Themes in Past Research

- Research on Safety Culture has been consistently high, but recent trends are beginning to critique it for being too narrow to explain accidents in complex socio-technical systems.
- Research on risk assessment methods (primarily PRA and HRA) has remained very prevalent and largely unchanged.
- However, critics wanted measures that could account for dynamic changes in system state (such as Monte Carlo simulation and Fuzzy clustering)
- Some critics argued that HRA methods should be used in connection with HFE analyses such as Task Analysis, Contextual Inquiry, Interviews, etc.

# Emerging Trends in HFE

## Resilience Engineering and Safety-II

- Resilience Engineering and Safety-II are gaining traction.
- **Organizational resilience** is growing, perhaps triggered by the Fukushima incident in 2011.
    - This can be used to supplement traditional approaches
    - Acknowledges that system functions are naturally interconnected, and these connections affect the system's ability to respond to extreme conditions.
- **Safety-II** focuses on learning from both success and failure.
    - The theory is well-established, but practical tools for Safety-II are still developing.
    - However, research in 2017 suggests it is more effect for describing human factors issues in operational events than traditional methods.

# Connections to Other Safety-Critical Industries

Construction, Rail, Healthcare, and Aviation

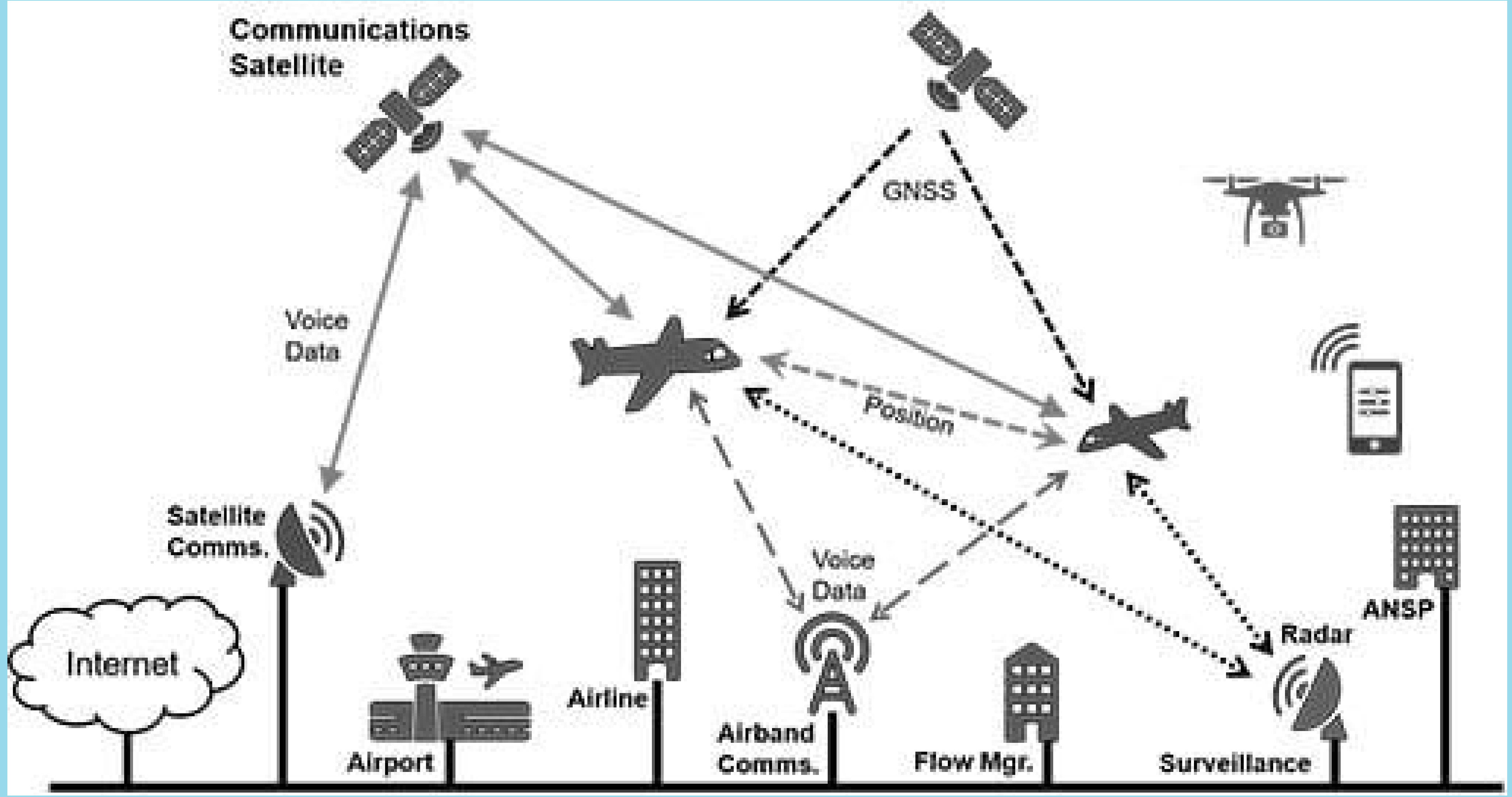
# Connections to Other Industries

- So far, healthcare has already begun implementing Safety-II for studying the blood transfusion process
- Researchers in Construction advocated for adaptive safety (which including empowering employees on the front line) to build organizational resilience.

# Current Best Tool for Safety-II and Resilience Engineering (RE)

## FRAM (Functional Resonance Analysis Method)

- Accounts for the variability in human performance within a system
- FRAM is a more established method which can be utilized in a safety-II perspective (focusing on improving performance, not just preventing bad performance).
- FRAM is more prevalent in other industries besides Nuclear.
- FRAM helped researchers in Aviation analyze 4 mid-air collisions to investigate the Air Traffic Management System (ATM) and found that it was unable to cope with adequately controlling aircraft.
- Other researchers integrated a Monte Carlo based framework with FRAM to provide a semi-quantitative format.



# FRAM and Non-Technical Skills (NTS)

- FRAM has helped other industries (starting with Aviation) to understand how non-technical skills (NTS) are critical for safety and success.
- Some important NTS include:
  - Decision making
  - Situation Assessment
  - Communication
  - Teamwork
  - Stress Management
- For managers, a critical NTS includes Safety Intelligence, which was  proposed in 2008) as an alternative to safety culture and "a way of helping top-level management understand safety and react appropriately, rather than just giving 'lip service.'"

# Other Resilience Engineering Research

## “Work as Done” vs “Work as Prescribed”

- This research finds that while procedures are important, they can be detrimental if they are too strict and/or don't account well for all possible scenarios.
- Instead, many studies have demonstrated the value of adaptations to work procedures that allow operators to have a more significant role in the process of rulemaking and outlining procedures.

# Future Challenges for HFE in the Nuclear Industry

# Few Attempts to Develop New Methods

- The literature review found that the Nuclear Industry has had very few attempts to develop new HFE methods or examining well-established constructs. This may be problematic, especially if the tools aren't well fit for the purpose for which they're being used.
- New technologies may pose significant problems with this.
- One of the major challenges with this might be a negative feedback loop driven by tight regulations:

“The day-to-day work of safety practitioners is heavily constrained by organizational objectives and directives (Provan et al., 2017). The safety management systems of organizations are responsive to legislative and regulatory requirements rather than evidence of what works.
“Regulators are waiting on researchers to tell them how to incorporate new theory into regulatory practice, and in the absence of this guidance have no choice but to base their rules on political priorities.
“Researchers can only conduct real-world research that organizations are willing to fund and engage with (Lamm, 2014). Each party blames the others for the lack of empirical research and evidence-based practice” (Rae et al., 2020)

# Conclusions

# Conclusions

- "Safety-II and resilience engineering [are gaining traction], albeit slowly."
- Safety-II and FRAM tools should be more thoroughly explored by practitioners.
- "Future research should focus on applying fresh thinking to the development of new HFE approaches catered for use by practitioners in the industry to increase safety and performance"
- One flaw with this research is that it does not include papers produced by nuclear regulators and operators.
    - So... Roger, what say you? Have these factors already been heavily researched at INL? Are there additional elements that I have not covered here? Are there topics discussed here that have not been addressed by INL?

# Questions?

Open floor for questions and discussion